

SOFTWARE ENGINEERING

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- Problem Statement:

People take hundred of photos and videos every year but often spend more time documenting their lives than experiencing them. Existing gallery and social media applications allow users to immediately view, edit, delete and repeatedly revisit their captured memories.

The proposed application will provide a different experience. Users will be limited to three photos per day. After capturing the memory, the user must lock it for a specific period of time which could be a month & a year. Once locked, the memory cannot be viewed until the selected period ends.

At the end of the year, the system will create a personalized yearly memory recap.

- Users:-

- Users who want to preserve meaningful memories
- who enjoy documenting their daily life.
- who wants a yearly digital album.

- Stakeholders:-

- Users - Captures and stores memories
- Administrator - Manages the platform and user accounts
- Developers - Develop and maintain the application
- Cloud/storage providers - securely stores photographs
- AI/API providers - Provides optional AI features

- Functional Requirements:-

→ User registration & login:

- User should be able to create an account using their basic information.
- User should be able to login & out easily.
- The system should provide password security.
- Each user's memory should be accessible only through their authenticated account.

→ Memory Capture:-

- User should be able to capture a photo directly through the application.
- The system should allow maximum three pictures per day.
- The daily limit should automatically reset after 24 hours.
- After capturing, the user should not be able to access it.

Memory Lock & Timer:-

After capturing a memory, the user should select a lock duration.

The available lock periods should be:

1 month, six months, 1 year.

The system should automatically calculate the unlock date.

Lock memories should remain inaccessible until their unlock date.

When the timer expires, the memory should be automatically available to the user.

Users should not be able to unlock a memory manually.

Non-Functional Requirements:-

Performance:

Photos/videos should upload quickly.

The lock system should work properly.

The recap should be generated within time.

Usability:

The interface should be simple.

Lock duration easy to select.

Risks:-

API Dependency:-

The application may depend on cloud storage, AI, camera, notification or other external API's. If these services become unavailable.

Deployment issue.

Large photo files requires significant storage & bandwidth. Deployment may become difficult if the hosting provider has limited storage.

Quality Attributes:

Security, Reliability, Privacy, Performance, Usability, Scalability, Maintainability, Availability.

Where AI is used:-

- AI can recognize whether it is a family picture, or with friends and organize it accordingly.
- It can also generate captions related with pictures.

Used Technology:-

Frontend:

HTML, CSS, Javascript, React.js.

Backend:

Python, Node.js + Express.

Database:

MySQL

APIs:

AI API, Cloud storage API, Camera API.

Hosting:

Vercel, Render, Firebase, AWS.

Current Limitation:

- Users are limited to 3 pictures.
- The system requires internet access for cloud storage.
- AI may incorrectly categorize some memories.
- If a user loses access to their account it will be hard to recover.